

SupplierPulse

Live Vendor Risk & Opportunity Intelligence Agent

A multi-agent telemetry orchestrator conducting real-time web scraping, signal classification, and alert automation for continuous supply chain visibility.

Lablab.ai Hackathon Submission | Built with Gemini API & Node.js

The Vulnerability of Stale Vendor Data

The Legacy Approach

Static Assessments: Relies on stale yearly audits, self-reporting questionnaires, and credit bureau scores.

Blindspots: Incidents like layoffs, leadership changes, or security breaches go unnoticed for months.

Delayed Action: Inability to react before a supplier experiences complete operational failure or bankruptcy.

Supply Chain Volatility

Rapid Shifts: Digital footprints and vendor risk indicators change in a matter of minutes.

Public Consensus Shift: Employee morale, customer complaints, and traffic dips precede financial failure.

Critical Threat: Modern enterprises face high dependencies on third-party APIs and suppliers.

Continuous Live Risk Telemetry

SupplierPulse Architecture

SupplierPulse acts as a 24/7 web intelligence crawler that automatically maps and monitors your suppliers. By utilizing multi-agent coordination, it evaluates live text data to detect early warning indicators.

1. Live Web Crawling

Scrapes 6 major digital vectors (Jobs, News, Filings, Reviews, Web, Social) using Search Grounding or Bright Data integration.

2. AI Signal Analysis

Classifies findings into 9 key signal categories, calculating severity and confidence indexes using AI agent inference.

3. Direct Integrations

Pushes summaries and alerts directly to Slack and Telegram channels on automated background monitoring cron cycles.

Multi-Agent Orchestration Flow

STEP 1

Phase 1: Input Normalization

Standardizes name variables and analyzes core industries (e.g. mapping 'Stripe' to Stripe, Inc.).

STEP 2

Phase 2: Multi-Vector Scraping

Triggers parallel ground-truth searches or API crawls for the target supplier name across web resources.

STEP 3

Phase 3: Telemetry Classification

AI models evaluate text blocks to score risks (severity, confidence) and extract key events.

STEP 4

Phase 4: Brief Synthesis

Compiles details into an executive summary, highlights key risks, and assigns a risk index (0-100).

The 6 Dimensions of Live Scraping



Jobs

Tracking hiring trends and layoffs to assess operational growth or cutbacks.



News

Scanning legal suits, funding rounds, regulatory compliance, and PR incidents.



Filings

Auditing official SEC regulatory reports (8-K, 10-Q) for corporate compliance.



Reviews

Scraping employee reviews (Glassdoor) to evaluate internal leadership and culture.



Web

Evaluating website uptime, traffic metrics, and domain authority indicators.



Social

Parsing public consensus and outage reports on Reddit and X/Twitter.

Real-Time UI & Alerting Integrations

Visual Risk Control Center

SSE Progress Streaming: Watch the multi-agent crawler execute in real-time as it scans sources.

Interactive Briefing Card: Drill down into individual risk signals, categorized with severity and confidence.

Historical Archives: Track a supplier's risk score over time to detect negative trajectories early.

Instant Incident Notifications

Slack Webhook: Richly-formatted briefs covering summaries and action plans directly inside Slack channels.

Telegram API: Instant alert notifications dispatched directly to telegram chat groups.

Chrono Monitor: Configure interval tasks (e.g. check vendor hourly/daily) to auto-trigger notifications.

Addressing a High-Growth Global Market

Total Addressable Market (TAM)

\$16.8 Billion

Global market spend for Governance, Risk, and Compliance (GRC) software.

Serviceable Addressable Market (SAM)

\$4.2 Billion

Enterprise spend focused specifically on vendor, supplier, and third-party risk analysis.

Target Customer Archetypes

Procurement Teams: Organizations managing large physical and digital supply chains.

Risk & Compliance Officers: Financial and corporate entities requiring strict vendor vetting for regulatory audits.

IT Security Leaders (CISOs): Technology teams seeking to track third-party SaaS outage history and security leaks.

Monetization & Commercial Strategy



SaaS Subscriptions

Tiered licensing packages (Professional, Enterprise) based on the number of active monitored vendor profiles.



API-as-a-Service

Usage-based billing allowing corporate customers to inject live risk scores directly into existing ERP and CRM software.



Custom Integrations

Service packages for setting up proprietary monitoring systems, dedicated databases, and high-frequency webhook pipelines.

The SupplierPulse Advantage

Comparing Vendor Vetting Systems

Feature	Legacy Credit Bureaus	SupplierPulse Agent
Analysis Frequency	Annual or monthly updates	Continuous real-time web crawlers
Signal Scopes	Financial spreadsheets & credit scores	6 vectors (Jobs, News, Social sentiment, etc.)
Notification Loop	Manual dashboard logins	Automated Slack & Telegram alerts
Risk Response Speed	Weeks to update database after bankruptcy	Identifies sentiment shifts in minutes

Future Outlook & Scalability Roadmap



Predictive Risk Index

Incorporate ML models to forecast supply disruptions by looking at correlated signals across similar vendors.



ERP Integration Plugins

Develop plug-and-play modules for SAP Ariba, Workday, and Salesforce for automatic vendor lockouts on high risk alerts.



AI Conversational Agent

Build a natural language procurement assistant enabling teams to query 'Summarize Stripe's latest compliance risk'.

Project Submission Checklist

Project Metadata

Project Title: SupplierPulse

Short Description: Live Vendor Risk & Opportunity Intelligence Agent offering real-time scraping, classification, and analysis.

Primary Categories: Multi-Agent Systems, Web Scraping, Risk Mitigation, Operations Automation.

Core Tech Stack: Node.js, Express, Vite, React, TypeScript, Gemini API (Search Grounding), Bright Data API, Slack & Telegram APIs.

Deliverables & Links

- ✓ Public GitHub Repository: Fully formatted and commented source code.
- ✓ Live Demo Application: Hosted Express/Vite prototype on web servers.
- ✓ Presentation Slides: Programmatically compiled PowerPoint and PDF presentation.